Table 2.2 An example of what current pump maintenance decisions will be improved and how.

Current decision	Improved decision via digital twin
Time-Based: If 6 months have passed, replace the bearing	**Predictive**: When trends indicate an 80% probability of failure within 4 weeks, order a bearing and schedule replacement in 3 weeks
Fixed Schedule: Maintain equipment based on a rigid schedule	**Risk-Based**: Optimize resource allocation by prioritizing maintenance on equipment with the highest risk of failure

2.1.4 Define a minimum viable digital representation

Based on decision-making requirements, determine the simplest digital representation that enables those improved decisions. Identify which physical attributes, behaviors, and relationships are essential and which can be simplified or omitted. For the pump example, the minimum viable digital representation requires capturing:

1. Vibration frequency spectrum (to detect bearing wear).
2. Temperature of bearing housings (to detect friction-induced heat).
3. Pump motor current signature (to detect mechanical load changes).
4. Operating hours and duty cycles (to understand usage).

This focused approach ensures every aspect of the digital representation serves a clear business purpose and can be validated against measurable results. Only after establishing these foundations should you proceed to capturing and modeling your physical systems.

2.2 A digital twin of the home

Let’s take the concepts introduced for defining your objectives, and create a practical worksheet that can be used to effectively map a physical system to its digital representation. To illustrate these concepts, consider the example of creating a digital twin of a home. For someone like myself, living in one of the driest cities in the world with moderately high electricity prices, the